

1. Charge

Build an electroscope

Two kinds of it, and it never appears from nowhere

Lesson 1 · about 90 minutes · everyone builds; no electrical hazard

What you need

- Sticky tape (regular, not invisible)
- Two balloons, string
- A glass jar with a plastic or cork lid
- Stiff copper wire, about 20 cm
- Aluminium foil
- A plastic comb or PVC pipe offcut
- A wool sweater or a silk scarf
- An empty aluminium can

Predict first

Rub a balloon on your hair and hold it near a thin stream of water from a tap. Does the stream bend *toward* the balloon, *away* from it, or not at all? Write down which, and one sentence saying why. Do this before you touch the tap.

Do it

1. **Two kinds.** Stick two 15 cm strips of tape onto the bench, then pull both off quickly and hang them near each other. They push apart. Now stick one strip *on top of* another, pull the pair off together, then peel them apart — now they pull together. Two strips pulled the same way repel; two pulled apart from each other attract. There are exactly two kinds of charge, and that is an experimental result, not a definition.
2. **Test the water.** Run the tap to a thin stream. Charge a balloon on hair or wool and bring it close. Compare with your prediction.
3. **Roll the can.** Lay the empty can on its side on a flat table. Bring the charged balloon near without touching. The can follows it.
4. **Build the electroscope.** Push the copper wire through the lid. Bend the inside end into a small hook and hang two strips of foil (about 1×4 cm) over it so they hang flat against each other. Crumple a ball of foil onto the outside end. Seal the lid onto the jar.
5. **Use it.** Charge the comb on wool and touch it to the foil ball. The leaves fly apart and *stay* apart. Touch the ball with your finger: they collapse.
6. **Charge without touching.** Recharge the comb. Bring it *near* the ball without contact — the leaves separate. Take it away — they fall. Now bring it near again, and while it is near, touch the ball with a finger from your other hand. Remove your finger first, then the comb. The leaves stay apart, and the electroscope now holds the *opposite* charge to the comb.

What it means

Two signs	Charge comes in two kinds, arbitrarily named positive and negative. Like repels like; opposite attracts. Everything else in this course is a consequence.
Rubbing moves, it does not create	Wool gives up electrons and the balloon takes them. The wool is left as positive as the balloon is negative. Nothing was made; something was moved. That is conservation of charge, and it has no known exceptions.
Why neutral things are attracted	The water and the can are neutral, and they still move. A nearby charge pulls their own charges slightly apart — the near side becomes opposite, the far side like. The near side is closer, so attraction wins. This is induction , and it is why the last step works.
The leaves	Both leaves get the same charge, so they repel. The angle is a crude voltmeter: it measures how much charge is on the ball.
Conductor and insulator	Your finger drained the charge to earth through your body; the glass jar does not. That difference is the whole reason wires and insulation both exist.

Break it

Humidity	Try this on a humid day and again on a dry one. Water in the air conducts, and charge leaks away. If nothing works, that is why — and it is also why the Tesla coil will perform differently on different days.
Longer leaves	Make one leaf twice as long. Does it move further? Why might that be about weight rather than charge?
Reverse it	Charge the comb with silk instead of wool. Does the electroscope respond the same way? Is the comb's charge the same sign?

The link to the coil
The topload of a Tesla coil is a large smooth conductor that holds charge, and a streamer is that charge leaving it into the air. The electroscope you just built is the same physics at a voltage a thousand times smaller. When the coil runs, you are watching your foil leaves fly apart — violently, sixty times a second.

This lesson is low voltage. Everything here runs from batteries or a small plug-in supply and cannot hurt you. That changes at Lesson 10. Build the habits now — one hand, discharge before touching, check before you power up — while the stakes are nothing, so that they are automatic when the stakes are real.

Telos. Electricity & Magnetism — 1. Charge. Revised 2026-07-25. Project-created text and diagrams are licensed CC BY 4.0. Incorporated public-domain artwork remains public domain and is credited on the plate it appears with. See LICENSE and CREDITS.md in the source. Nothing here is professional advice; verify current regulations and follow the stated safety procedure.